Three-Qubit Gates, Inner Products, Entanglement and Bell States

EL4043 – Teknologi dan Komputasi Kuantum

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STEI, ITB

October 2, 2026

Lecture Roadmap

1. Three-qubit gates
2. Inner products
3. Self-inner products and normalization
4. Entanglement
5. Product versus entangled states
6. Bell states
7. Basis-dependent correlations
8. Bell and CHSH concepts
9. Qiskit and Quirk demonstrations

Main idea

We will move from the mathematics of quantum states and gates toward experimentally visible quantum correlations.

1. Three-Qubit Gates

Extending controlled operations beyond two qubits.

Why Three-Qubit Gates?

For two qubits we already have gates such as

$$X, \qquad H, \qquad \text{CNOT}.$$

For larger quantum circuits, we need operations involving three or more qubits. Examples include:

- Toffoli gate
- Fredkin gate
- controlled-controlled-Z
- multi-controlled gates

A three-qubit system has Hilbert-space dimension

$$\boxed{2^3 = 8.}$$

The Toffoli Gate

The Toffoli gate is also called

$$\boxed{\text{CCNOT}}.$$

It contains:

- two control qubits,
- one target qubit.

The target flips only when both controls are 1. Therefore,

$$\boxed{|c_1 c_2 t\rangle \rightarrow |c_1 c_2,\ t \oplus (c_1 \wedge c_2)\rangle}$$

where $\wedge$ means logical AND.

Toffoli Truth Table

Input	Output
000	000
001	001
010	010
011	011
100	100
101	101
110	111
111	110

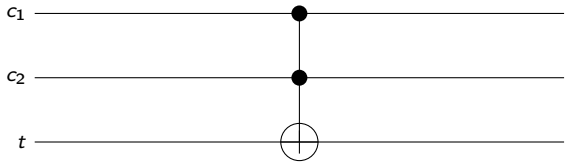
Only when both controls are 1 does the target flip:

$$\boxed{110 \rightarrow 111}$$

and

$$\boxed{111 \rightarrow 110.}$$

Toffoli Circuit



The two black dots are controls. The $\oplus$ symbol is the target.

Target flips only when both controls are 1.

Toffoli in Qiskit

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(3)
qc.ccx(0, 1, 2)

print(qc)
```

The command

`qc.ccx(0,1,2)`

means:

- q_0: first control
- q_1: second control
- q_2: target

For the input

$$|110\rangle,$$

the output is

$$\boxed{|111\rangle}.$$

2. Inner Products

Understanding overlap, orthogonality and normalization.

What Is an Inner Product?

For two quantum states

$$|\psi\rangle \qquad \text{and} \qquad |\phi\rangle\,,$$

the inner product is

$$\boxed{\langle\phi|\psi\rangle\,.}$$

It produces a complex number:

$$\langle\phi|\psi\rangle \in \mathbb{C}.$$

It describes the mathematical overlap between the two states.

Bra and Ket

Suppose

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

The corresponding bra is the conjugate transpose:

$$\boxed{\langle\psi| = \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}.}$$

Therefore,

$$\boxed{\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*.}$$

The complex conjugation is essential.

Example: $\langle 0|1\rangle$

We have

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \langle 0|1\rangle &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \boxed{0}. \end{aligned}$$

Thus,

$$\boxed{|0\rangle \perp |1\rangle.}$$

The computational basis states are orthogonal.

Self-Inner Product

The inner product of a state with itself is

$$\boxed{\langle\psi|\psi\rangle\,.}$$

For

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,,$$

we obtain

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2.$$

A normalized quantum state satisfies

$$\boxed{\langle\psi|\psi\rangle = 1.}$$

Self-Inner Product and Norm

The squared norm of a state is

$$\boxed{\|\psi\|^2 = \langle\psi|\psi\rangle\,.}$$

For example,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Then

$$\begin{aligned}\langle+|+\rangle &= \frac{1}{2}(\langle 0| + \langle 1|)(|0\rangle + |1\rangle) \\ &= \frac{1}{2}(1+1) \\ &= \boxed{1}.\end{aligned}$$

Therefore $|+\rangle$ is normalized.

Orthogonal Superposition States

Consider

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

and

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Then

$$\begin{aligned}\langle +|-\rangle &= \frac{1}{2}(\langle 0| + \langle 1|)(|0\rangle - |1\rangle) \\ &= \frac{1}{2}(1-1) \\ &= \boxed{0}.\end{aligned}$$

Thus,

$$\boxed{|+\rangle \perp |-\rangle .}$$

3. Entanglement and Bell States

From product states to genuinely quantum correlations.

3.1 Product States

A two-qubit state is a product state if it can be written as

$$\boxed{|\psi\rangle_{AB} = |\phi\rangle_A \otimes |\chi\rangle_B .}$$

Example:

$$|+0\rangle = |+\rangle \otimes |0\rangle .$$

Expanding,

$$\boxed{|+0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}.}$$

Therefore this state is separable.

3.1 Entangled States

Consider

$$\boxed{|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.}$$

There are no single-qubit states $|\phi\rangle$ and $|\chi\rangle$ such that

$$|\Phi^+\rangle = |\phi\rangle \otimes |\chi\rangle .$$

Therefore,

$$\boxed{|\Phi^+\rangle \text{ is entangled.}}$$

Simple distinction

Product: factorizable into subsystem states. Entangled: cannot be factorized into subsystem states.

Product vs Entangled: Simple Comparison

Product

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$$= |+\rangle \otimes |0\rangle$$

Therefore:

Product

Entangled

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Cannot be factorized. Therefore:

Entangled

Superposition does not automatically mean entanglement.

3.2 The Four Bell States

The four Bell states are

$$\boxed{\begin{aligned} |\Phi^+\rangle &= \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \\ |\Phi^-\rangle &= \frac{|00\rangle - |11\rangle}{\sqrt{2}}, \\ |\Psi^+\rangle &= \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \\ |\Psi^-\rangle &= \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \end{aligned}}$$

They form an orthonormal basis of the two-qubit Hilbert space.

Bell States and Relative Phase

Compare

$$\left|\Phi^{+}\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

with

$$\left|\Phi^{-}\right\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}.$$

The computational-basis probabilities are identical:

$$P(00) = P(11) = \frac{1}{2}.$$

However, the relative phase is different. This difference becomes observable when we measure in an appropriate different basis.

Key point

Relative phase can affect interference and correlations even when computational-basis probabilities are unchanged.

Creating $|\Phi^+\rangle$

Start with

$$|00\rangle\,.$$

Apply H to the first qubit:

$$|00\rangle \to \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

Then apply CNOT:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}} \to \boxed{\frac{|00\rangle + |11\rangle}{\sqrt{2}}}.$$

Therefore,

$$\boxed{|00\rangle \xrightarrow{H} \xrightarrow{\text{CNOT}} |\Phi^+\rangle\,.}$$

Generating the Other Bell States

Starting from

$$\left|\Phi^{+}\right\rangle,$$

local Pauli gates can generate the other Bell states:

$$\boxed{\begin{aligned}(I \otimes Z)\left|\Phi^{+}\right\rangle &= \left|\Phi^{-}\right\rangle,\\ (I \otimes X)\left|\Phi^{+}\right\rangle &= \left|\Psi^{+}\right\rangle.\end{aligned}}$$

Applying both X and Z gives $\left|\Psi^{-}\right\rangle$ up to an irrelevant global phase. Thus, all four Bell states are related by local operations.

3.3 Basis-Dependent Correlations

Consider

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

In the computational basis:

$$\boxed{P(00) = P(11) = \frac{1}{2}.}$$

Therefore the results are correlated. Now consider the X basis:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

The same state can be written as

$$\boxed{|\Phi^+\rangle = \frac{|{+}{+}\rangle + |{-}{-}\rangle}{\sqrt{2}}.}$$

The Same Bell State in Two Bases

For $|\Phi^+\rangle$:

$$\boxed{|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}}$$

in the Z basis. Therefore:

$$00, \quad 11.$$

In the X basis:

$$\boxed{|\Phi^+\rangle = \frac{|{+}{+}\rangle + |{-}{-}\rangle}{\sqrt{2}}}$$

Therefore:

$$++, \quad --.$$

Important

The quantum state is the same. The measurement basis has changed.

Changing the Measurement Basis

To measure a qubit in the X basis using a computational-basis measurement:

$$\boxed{H \rightarrow \text{computational-basis measurement.}}$$

For two qubits:

$$\boxed{H \otimes H \rightarrow \text{measure.}}$$

Thus the circuit

$$|\Phi^+\rangle \xrightarrow{H \otimes H} \text{measure}$$

measures the original Bell state in the X basis. This is extremely useful for demonstrating basis-dependent correlations in Qiskit.

3.4 Bell's Theorem: Basic Idea

Bell's theorem concerns whether certain quantum correlations can be explained by theories based on local hidden variables.
The conceptual comparison is:

Local hidden-variable model

versus

Quantum mechanics.

Bell-type inequalities provide quantitative constraints on the correlations allowed by the relevant class of local hidden-variable models.
Quantum mechanics predicts violations for suitable entangled states and measurement settings.

The CHSH Scenario

There are two parties:

$$\boxed{A \quad \text{and} \quad B.}$$

Each chooses between two measurement settings:

$$A_0, \quad A_1$$

and

$$B_0, \quad B_1.$$

Each measurement produces

$$\boxed{+1 \quad \text{or} \quad -1.}$$

We therefore obtain four correlation functions:

$$E(A_0, B_0), \quad E(A_0, B_1), \quad E(A_1, B_0), \quad E(A_1, B_1).$$

The CHSH Parameter

The CHSH quantity is

$$\boxed{S = E(A_0, B_0) + E(A_0, B_1) + E(A_1, B_0) - E(A_1, B_1).}$$

For local hidden-variable theories,

$$\boxed{|S| \leq 2.}$$

Quantum mechanics allows

$$\boxed{|S| \leq 2\sqrt{2}.}$$

The value

$$2\sqrt{2}$$

is known as the Tsirelson bound.

What Is a Correlation?

Suppose Alice obtains

$$a \in \{+1, -1\}$$

and Bob obtains

$$b \in \{+1, -1\}.$$

The correlation is

$$\boxed{E(A, B) = \langle ab \rangle.}$$

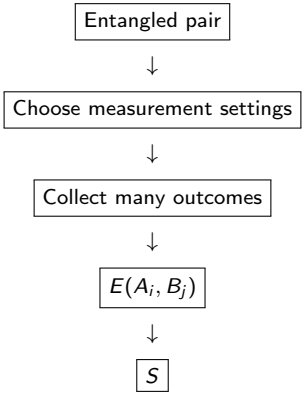
Interpretation:

$E(A, B)$	Interpretation
$+1$	perfect correlation
0	no linear correlation
-1	perfect anti-correlation

In an experiment, $E(A, B)$ is estimated from many measurement shots.

Bell/CHSH: The Conceptual Picture

The complete workflow is



The important distinction is:

ordinary correlation $\neq$ Bell-inequality violation.

5. Live Demonstrations

Quirk and Qiskit

Demonstration 1: Toffoli

Goal: demonstrate a three-qubit controlled gate.
Start with

$$|110\rangle\,.$$

Apply CCNOT:

$$\boxed{|110\rangle \to |111\rangle\,.}$$

Then change the input to

$$|100\rangle\,.$$

Now:

$$\boxed{|100\rangle \to |100\rangle\,.}$$

Question for YOU Guys

Why does the target flip in the first case but not the second?

Demonstration 1: Qiskit Code

```
from qiskit import QuantumCircuit
qc = QuantumCircuit(3)
qc.x(0)
qc.x(1)
qc.ccx(0, 1, 2)

print(qc)
```

The first two X gates prepare the controls in $|1\rangle$.
Then:

$$\boxed{\text{CCNOT} : 110 \rightarrow 111.}$$

Demonstration 2: Bell State

Prepare

$$\left|\Phi^{+}\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Circuit:

$$|00\rangle \xrightarrow{H} \xrightarrow{\text{CNOT}} \left|\Phi^{+}\right\rangle.$$

Expected ideal measurement:

$$\boxed{P(00) = \frac{1}{2}, \qquad P(11) = \frac{1}{2}.}$$

while

$$P(01) = P(10) = 0.$$

Demonstration 2: Qiskit Code

```
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator

qc = QuantumCircuit(2, 2)
qc.h(0)
qc.cx(0, 1) qc.measure([0, 1], [0, 1])

simulator = AerSimulator()
result = simulator.run( qc,
shots=1024 ).result()

print(result.get_counts())
```

Expected:

$$00 \approx 50\%, \qquad 11 \approx 50\%.$$

Demonstration 3: Product vs Entangled

First circuit:

$$|00\rangle \xrightarrow{H_0} \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

Expected outcomes:

$$00, \quad 10.$$

Second circuit:

$$|00\rangle \xrightarrow{H_0} \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Expected outcomes:

$$00, \quad 11.$$

Teaching point

Both states contain superposition. Only the second is entangled.

Demonstration 4: Measurement Basis

Prepare

$$|\Phi^+\rangle.$$

Measurement 1 Measure directly.

$$\boxed{00, \quad 11}$$

Measurement 2
Apply

$$H \otimes H$$

before measurement.
Now we are measuring in the X basis:

$$\boxed{++, \quad --}$$

Teaching point

The state did not change. The measurement basis changed.

Demonstration 4: Qiskit

```
qc = QuantumCircuit(2, 2)
# Prepare Bell state
qc.h(0)
qc.cx(0, 1)
# Change measurement basis
qc.h(0)
qc.h(1)
qc.measure([0, 1], [0, 1])
```

The two final Hadamard gates implement the basis transformation. This allows students to compare:

Z-basis measurement

with

X-basis measurement.

Demonstration 5: CHSH

A complete CHSH demonstration requires:

1. preparing an entangled pair,
2. choosing measurement settings,
3. repeating each setting many times,
4. estimating four correlations,
5. calculating S.

The workflow is

$$\boxed{\text{Circuit} \rightarrow \text{Counts} \rightarrow E(A_i, B_j) \rightarrow S.}$$

This is better suited to Qiskit than to a purely visual simulator.

Quirk vs Qiskit

Topic	Quirk	Qiskit
Toffoli	Excellent	Excellent
Bell state	Excellent	Excellent
Product vs entangled	Excellent	Excellent
Measurement probabilities	Excellent	Excellent
Inner products	Limited	Good
Basis changes	Excellent	Excellent
Bell correlations	Good	Excellent
CHSH calculation	Limited	Excellent

Simple strategy

Use **Quirk for visual intuition** and **Qiskit for quantitative experiments**.

Class Exercise: Inner Product

Consider

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

and

$$|\phi\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Calculate

$$\boxed{\langle\psi|\phi\rangle}$$

and

$$\boxed{\langle\psi|\psi\rangle\,.}$$

Expected:

$$\boxed{\langle\psi|\phi\rangle = 0, \qquad \langle\psi|\psi\rangle = 1.}$$

Class Exercise: Product or Entangled?

Classify each state:

$$|\psi_1\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

and

$$|\psi_2\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Hint: Ask whether the state can be written as

$$|\phi\rangle \otimes |\chi\rangle\,.$$

Answers:

$\boxed{|\psi_1\rangle \text{ is product}}$

and

$\boxed{|\psi_2\rangle \text{ is entangled.}}$

Class Exercise: Toffoli

Start with

$$|110\rangle .$$

Apply CCNOT.
What is the output?

Then start with

$$|100\rangle .$$

What is the output?

Think

What condition must be satisfied before the target qubit flips?

Class Exercise: Bell State

Start with

$$|00\rangle .$$

Apply:

$$H$$

to the first qubit, followed by

$$\text{CNOT}.$$

Answer:

1. What is the final state?
2. What outcomes can be measured?
3. What is the probability of each?
4. Is the state product or entangled?

Expected final state:

$$\boxed{|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.}$$

Summary

1. The Toffoli gate is a three-qubit controlled operation.
2. Inner products quantify overlap between states.
3. $$\langle \psi | \psi \rangle = 1$$
   for a normalized state.
4. Product states can be factorized.
5. Entangled states cannot be factorized.
6. There are four Bell states.
7. The same Bell state can show correlations in different measurement bases.
8. CHSH combines four correlation measurements into one quantity.
9. Qiskit is useful for quantitative experiments; Quirk is useful for visual intuition.

Key Takeaways

Three qubits $\rightarrow$ Toffoli

$\langle|\phi|\psi\rangle \rightarrow$ overlap

$\langle\psi|\psi|\psi|\psi\rangle \rightarrow$ normalization

Bell states $\rightarrow$ entanglement

Measurement basis $\rightarrow$ observed correlations

CHSH $\rightarrow$ quantitative Bell test

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Thank You

Questions?

Three Qubits $\rightarrow$ Inner Products $\rightarrow$ Entanglement $\rightarrow$ Bell/CHSH